

Sprint 2: Flu Vaccination Predictor Project

By ALPHA GROUP

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 Based on CDC H1N1 2009 Survey Data

Project Overview

- **Goal:** Predict flu vaccination using CDC survey data (26,707 records)
- **Data Types:** Behavior, demographics, health beliefs, and access to care
- **Main Objectives:**
 - Identify factors affecting vaccine uptake
 - Build machine learning models
 - Recommend actions for public health

Compare Label vs One-Hot Encoding

Label Encoding

Final shape: (26707, 38)

Remaining nulls: 0

		precision	recall	f1-score	support
	0	0.87	0.94	0.91	4212
	1	0.70	0.48	0.57	1130
accuracy				0.85	5342
macro avg		0.78	0.71	0.74	5342
weighted avg		0.83	0.85	0.83	5342

One-Hot Encoding

Final shape: (26707, 38)

Remaining nulls: 0

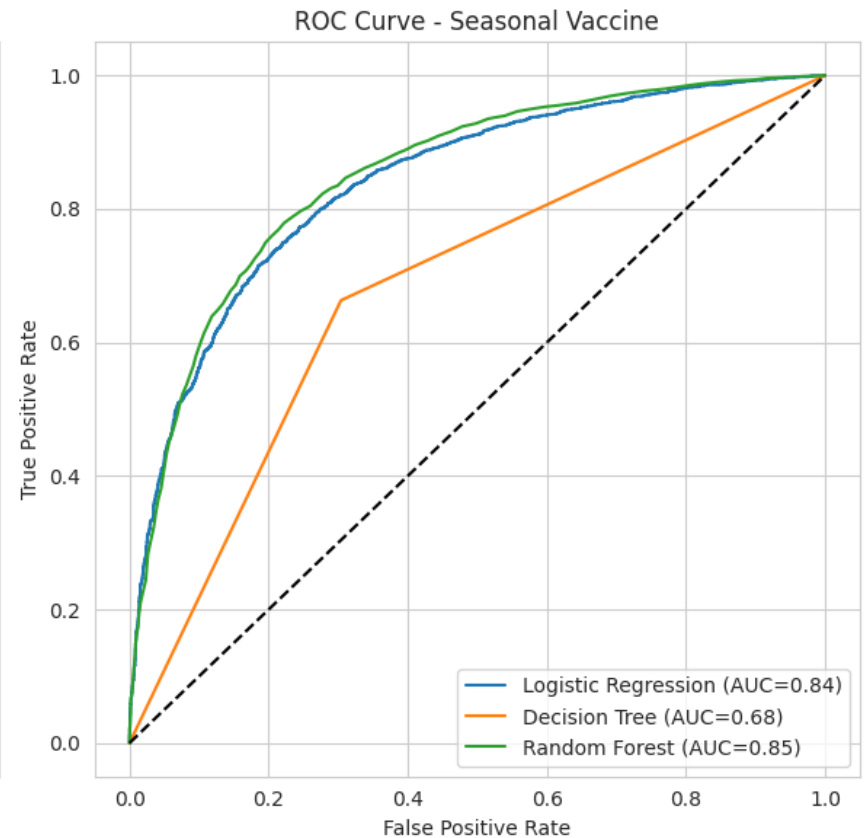
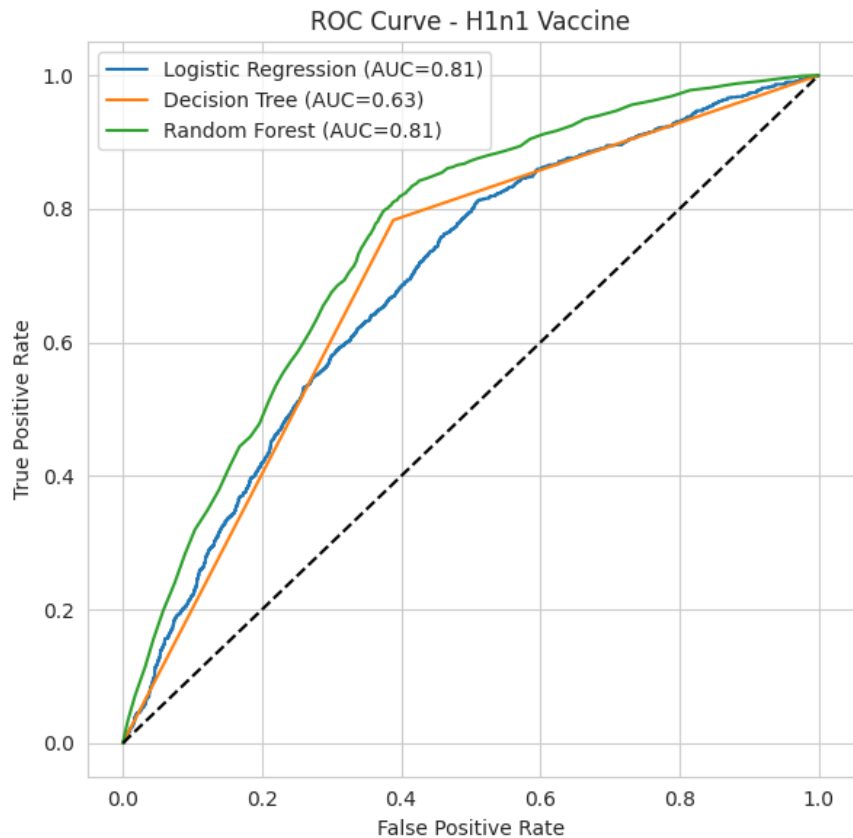
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Model Performance

Both encoding methods (Label vs One-Hot) gave identical results:

- Accuracy: 85%
- Prediction quality:
 - Excellent at identifying who won't vaccinate (94% correct)
 - Moderate at identifying who will vaccinate (48% correct)

Baseline Model Performance Comparison



The Model Comparator For H1N1 and Seasonal Results

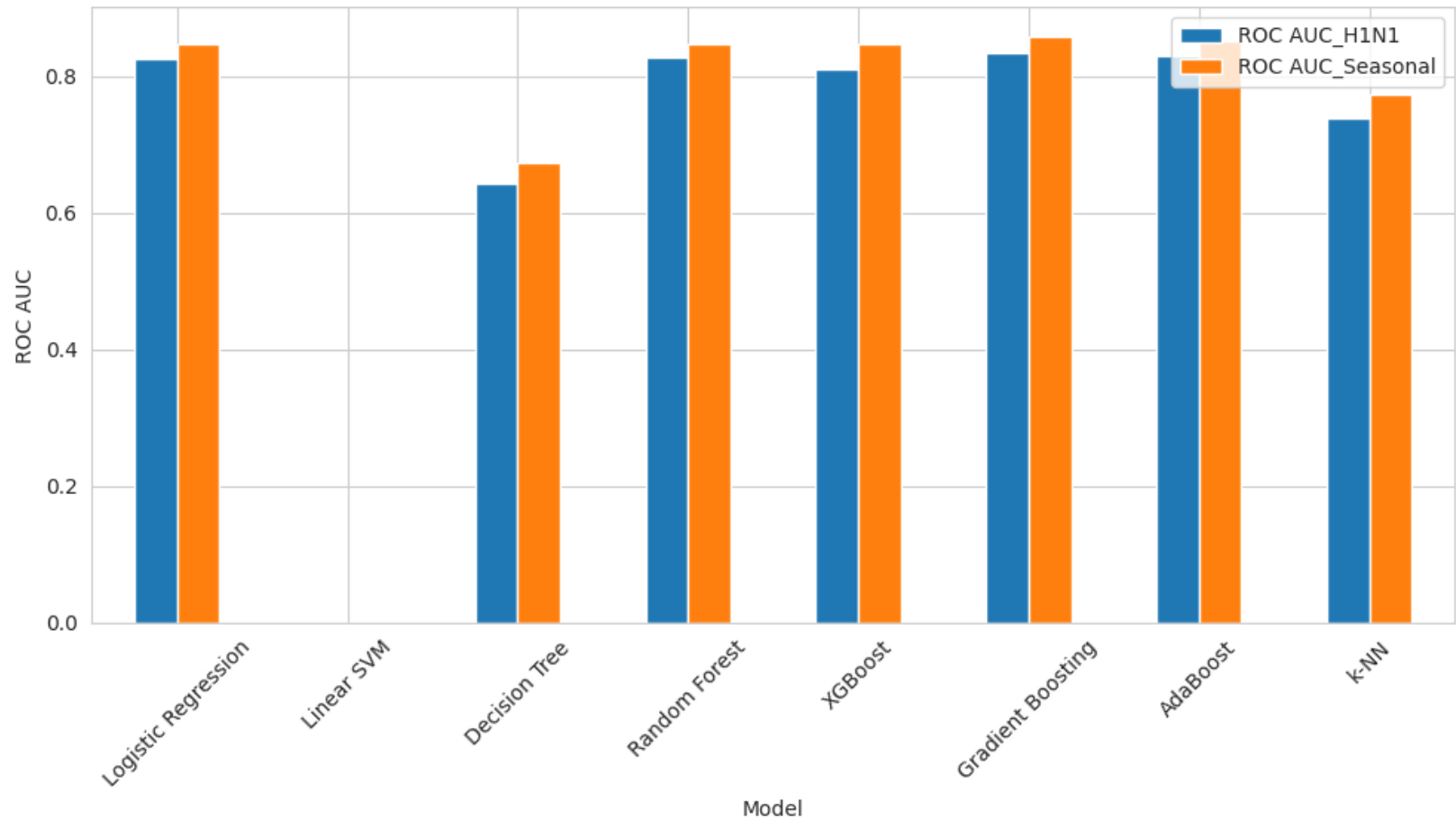
H1N1 Model Results:

	Model	Train Time (s)	Accuracy	Precision	Recall
	Gradient Boosting	3.079693	0.844440	0.713267	0.442478
	AdaBoost	0.688369	0.839573	0.703428	0.417699
	Random Forest	2.312499	0.839199	0.703148	0.415044
	Logistic Regression	0.100264	0.839760	0.703264	0.419469
	XGBoost	0.340557	0.831149	0.639364	0.462832
	k-NN	0.003762	0.813553	0.597668	0.362832
	Decision Tree	0.137255	0.754773	0.426350	0.461062
	Linear SVM	0.190614	0.831524	0.643750	0.455752

Seasonal Model Results:

	Model	Train Time (s)	Accuracy	Precision	Recall
	Gradient Boosting	3.084647	0.787158	0.773750	0.757650
	AdaBoost	0.674140	0.783227	0.777349	0.739290
	Random Forest	2.245748	0.781542	0.770430	0.746226
	Logistic Regression	0.156420	0.781355	0.773095	0.740922
	XGBoost	0.277709	0.778547	0.763071	0.750306
	k-NN	0.002722	0.718458	0.698532	0.679723
	Decision Tree	0.145816	0.676713	0.646677	0.651163
	Linear SVM	0.282451	0.773680	0.773087	0.717258

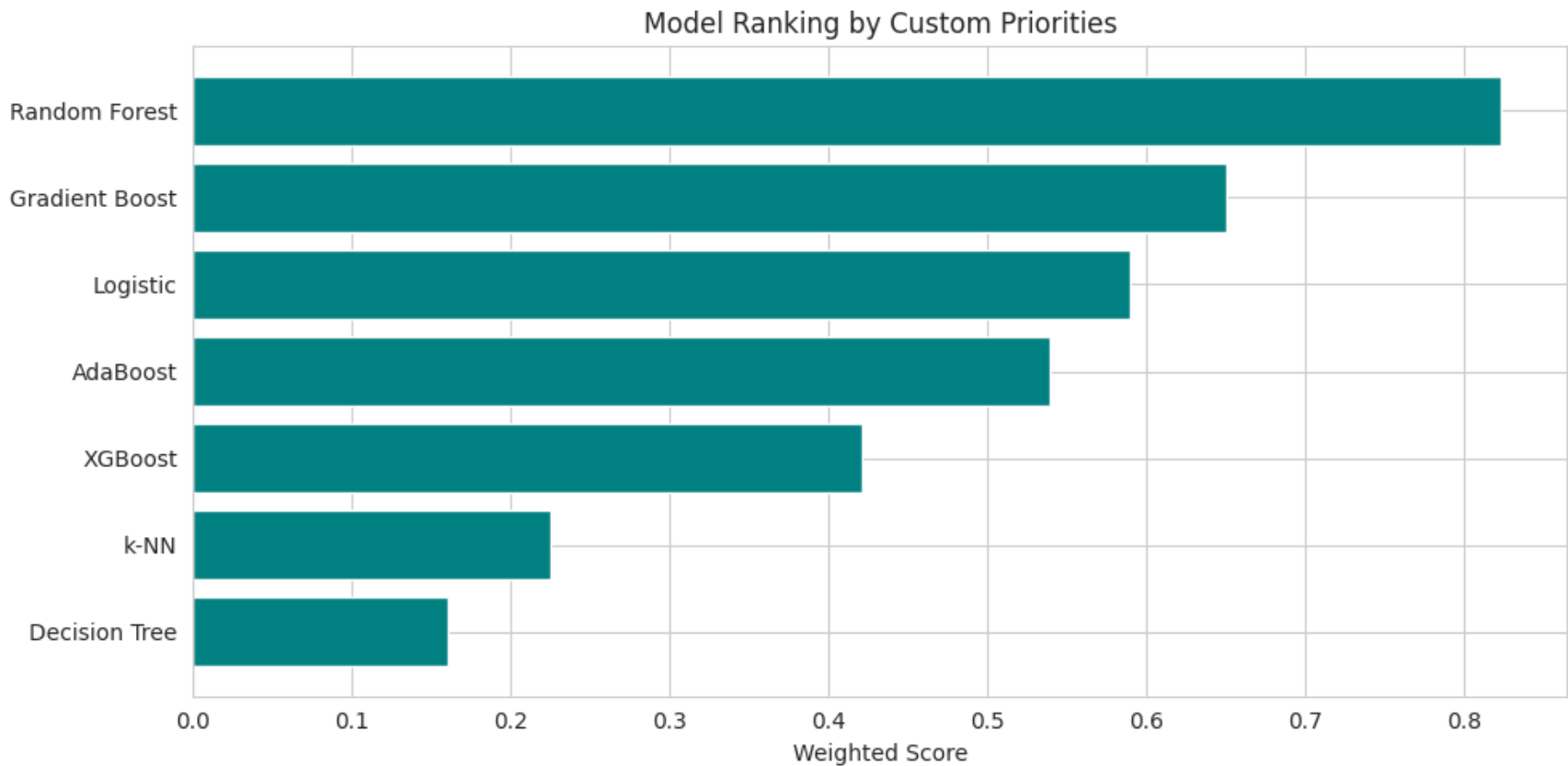
Model Performance Comparison Between Targets



Strengths & Weaknesses Of Models

Model	Strength	Weakness
Random Forest	High AUC, good overall balance	Higher memory usage
Gradient Boost	High performance and F1 score	Longer training time
Logistic	Very interpretable, fast, low memory	Slightly low recall and AUC

Best Models by Priority



Model Selection Guide

- Use Random Forest for real-world balance
- Use Gradient Boosting when accuracy is priority
- Use Logistic Regression when interpretability matters

Conclusion

- Gradient Boosting and Random Forest are top models
- Model insights can guide smarter public health actions
- Future work will focus on visualization and deployment